

Freight Rate Prediction

Machine Learning Engineer Assessment — Technical Report

TASK	MODEL	CV MAE	HOLDOUT MAE
Spot freight rate prediction	LightGBM 33 features	\$130.02	\$129.48

1. Executive Summary

The assessment is a supervised regression problem: predict **posted_rate** for future freight loads using historical January–October 2025 data.

- 48,000 labeled development loads are used for model development.
- 12,000 unlabeled November–December loads receive final predictions.
- A separate 31-day December benchmark covers a fixed Lexington, KY → Fort Wayne, IN lane.
- The final model is a LightGBM regressor using 33 engineered features and out-of-fold target encodings.
- All preprocessing and target statistics are isolated to the training data during validation.

2. Key Results

Evaluation	MAE	RMSE	R ²	MAPE	MedAE
CV Fold 1 — Aug	\$133.78	\$627.60	0.818913	6.59%	\$52.16
CV Fold 2 — Sep	\$126.22	\$617.54	0.835657	5.50%	\$46.89
CV Fold 3 — Oct	\$130.07	\$653.45	0.817247	6.32%	\$42.63
3-Fold Average	\$130.02	\$632.87	0.823939	6.14%	\$47.23
Sep–Oct Holdout	\$129.48	\$635.27	0.826706	5.96%	\$44.84

Scorer: score.py completed successfully with exit code 0; 12,000 validation predictions and 31 December predictions passed structural checks.

Important: these are internal development metrics. Spotter calculates the official validation score after submission.

3. Data and Preprocessing

The development data contains 48,000 historical loads from January 1 through October 31, 2025. The validation set contains 12,000 future loads from November 1 through December 31, 2025.

File	Rows	Period	Target
train-test.csv	48,000	Jan–Oct 2025	posted_rate
validation.csv	12,000	Nov–Dec 2025	Held out
december-chart-inputs.csv	31	Dec 2025	Benchmark

Data Quality

Issue	Scope	Handling
Negative weights	292 train rows	abs(weight)
Missing weights	300 train rows	Equipment-specific training median; global training fallback
Unseen validation cities	8 pickup + 8 delivery	Global training mean for target encoding; validation coordinates retained

All preprocessing estimators are fitted on the training partition and then frozen before validation or inference. The eight unseen validation cities retain their supplied geographic coordinates for spatial features.

4. Feature Engineering

Group	Count	Features
Distance	3	distance, log_distance, distance_sq
Weight	2	weight, log_weight
Equipment	2	equipment_code, equip_premium
Calendar	7	day_of_week, day_of_month, is_weekend, sin/cos encodings
Coordinates	4	pickup/delivery latitude and longitude
Spatial geometry	8	haversine, circuitry, coordinate deltas, midpoints
Interactions	3	distance × weight, distance × premium, log-distance × premium
Target encodings	4	pickup, delivery, equipment RPM, baseline rate

Target encoding uses 5-fold out-of-fold statistics with Bayesian smoothing ($m = 15$), so a training row does not use its own target when its encoding is calculated.

5. Model

The final estimator is a single LightGBM regressor configured for the tabular freight-rate problem. The final feature matrix contains 33 engineered variables.

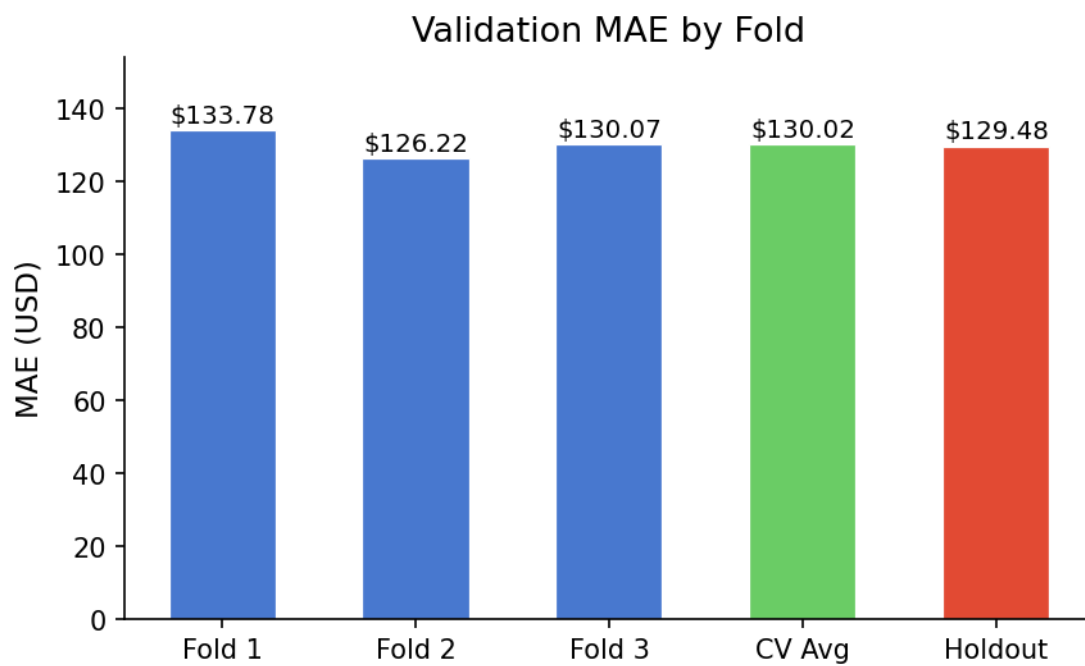
Parameter	Value
Objective / metric	regression / MAE
num_leaves	63
learning_rate	0.03
feature_fraction	0.85
bagging_fraction / frequency	0.85 / 5
min_child_samples	20
reg_alpha / reg_lambda	0.1 / 0.1
n_estimators	197
random_state	42

6. Time-Based Validation

Random splits were not used. The validation protocol follows the forward-looking nature of the task.

Split	Training	Validation
Fold 1	Jan–Jul 2025	Aug 2025
Fold 2	Jan–Aug 2025	Sep 2025
Fold 3	Jan–Sep 2025	Oct 2025
Holdout	Jan–Aug 2025	Sep–Oct 2025

For every split, imputation, coordinate mappings, and target encodings are fitted using only the training partition. The September–October holdout provides a contiguous two-month out-of-time check.



Validation MAE across the three chronological folds, their average, and the two-month holdout.

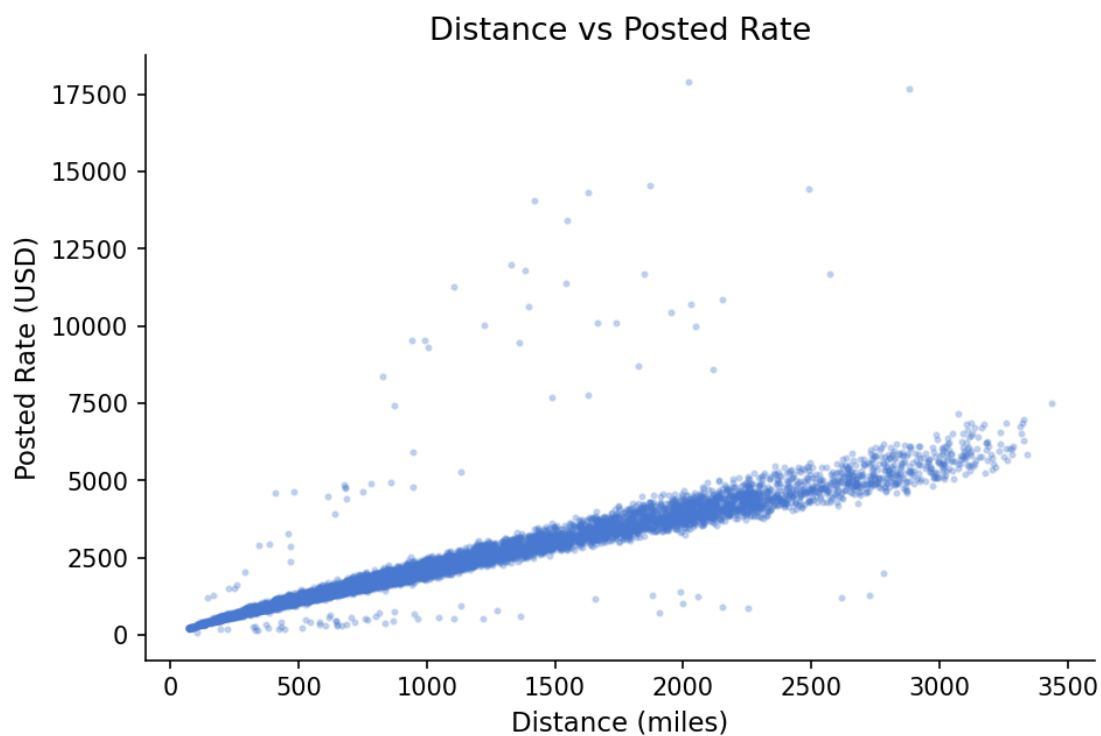
7. Model Experiments

Two auxiliary training fields, **market_index** and **quote_signal**, were tested through an ablation. Removing them reduced the average 3-fold CV MAE from **\$161.07** to **\$154.20**. They were excluded from the final architecture because their temporal behavior was not reliable for the out-of-time setup and they were not available in the December benchmark input schema.

Subsequent engineering changes—including out-of-fold target encoding, spatial geometry features, and the final tree configuration—produced the verified 3-fold average MAE of **\$130.02**.

8. Final Validation Results

Evaluation	MAE	RMSE	R ²	MAPE	MedAE
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Distance versus posted rate in the training data.

The MAE and RMSE differ substantially because the target contains a relatively small number of high-value observations. Development metrics should be interpreted as historical estimates rather than guarantees of future performance.

9. December Benchmark

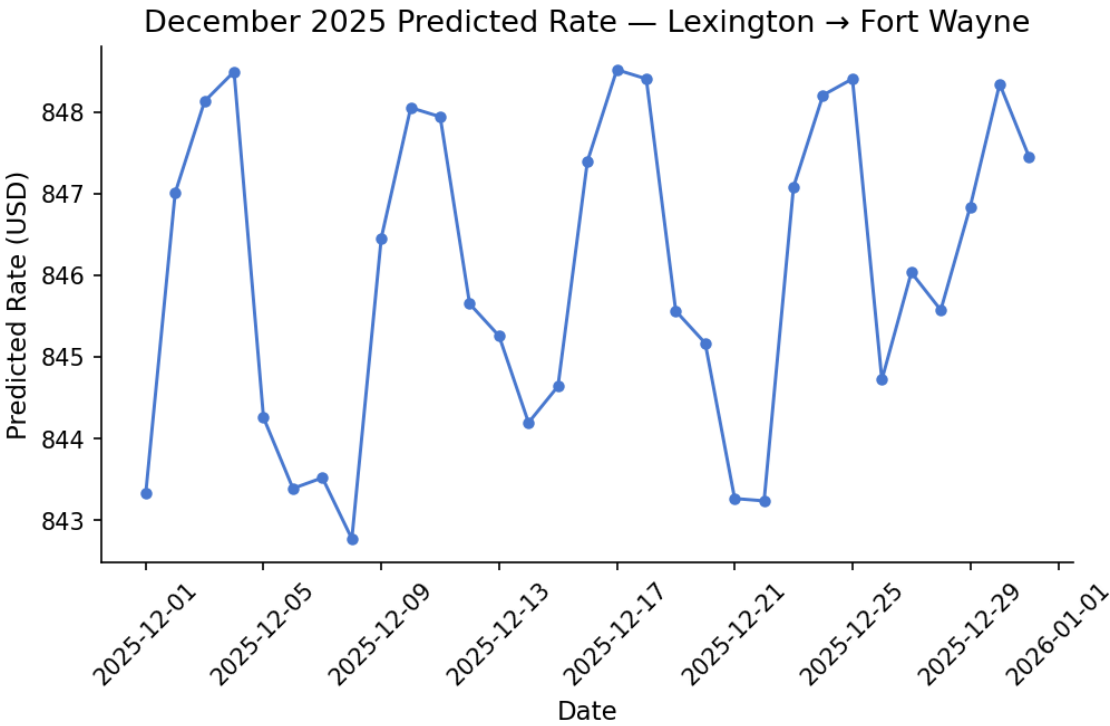
The December benchmark keeps the shipment configuration fixed and varies only the date.

Route	Lexington, KY → Fort Wayne, IN
Distance	360 miles
Equipment	Dry Van
Weight	32,000 lb
Dates	December 1–31, 2025

Minimum	Maximum	Mean	Spread
\$842.77	\$848.53	\$846.05	\$5.75

Because the route, equipment, and weight are fixed, the December variation comes from the date-dependent calendar features used by the model. The predictions are generated directly by the final model without post-hoc multipliers or synthetic seasonal overrides.

Training data ends on October 31, 2025, so specific Christmas and New Year's freight disruptions are not directly represented in training. The December output therefore reflects calendar patterns learned from January through October.



Final December predictions for the fixed benchmark corridor.

10. Submission Verification

Check	Result
Official score.py	Exit code 0
Validation predictions	12,000 rows; positive finite values; IDs/order validated
December predictions	31 rows; valid schema and finite values
December chart	Generated successfully by score.py

11. Limitations

- Official validation performance is determined by Spotter after submission against held-out labels.
- Development data ends on October 31, 2025.
- December-specific holiday disruptions are not represented in the training period.
- market_index and quote_signal are excluded from the final architecture.

12. Reproducibility

Place the provided assessment CSV files in the repository root, install the dependencies, and run the training pipeline:

```
pip install -r requirements.txt
```

```
python train_model.py
```

The standalone scorer can be run with:

```
python score.py --predictions validation_predictions.csv --december-predictions  
december_predictions.csv
```

13. Final Deliverables

- validation_predictions.csv — 12,000 final validation predictions.
- december_predictions.csv — 31 December benchmark predictions.
- score.py — official structural verification script.
- Spotter_ML_Assessment_Report.pdf — this technical report.

Final development benchmark	MAE \$130.02	Holdout MAE \$129.48
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